



3.7 Lines and planes in three dimensions (HL)

Geometry and trigonometry · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A line in 3D has vector equation $\mathbf{r} = \mathbf{a} + t\mathbf{b}$, where $\mathbf{a}$ is a point on the line and $\mathbf{b}$ a direction vector. A plane has equation $\mathbf{n} \cdot \mathbf{r} = \mathbf{n} \cdot \mathbf{a}$ (or $ax+by+cz=d$), where $\mathbf{n}$ is a normal vector and $\mathbf{a}$ a point on the plane. The angle between two lines uses the dot product of direction vectors; the angle between two planes uses the dot product of normal vectors; the angle between a line and a plane uses 90° minus the angle between the line's direction and the plane's normal. Intersections are found by substitution or by solving simultaneous equations.

Worked example: follow the reasoning

Find the point of intersection of the line $\mathbf{r}=(1,0,2)+t(1,1,-1)$ and the plane $x+2y-z=3$.

Substitute: $x=1+t$, $y=t$, $z=2-t$.

$(1+t)+2(t)-(2-t)=3 \rightarrow 1+t+2t-2+t=3 \rightarrow 4t-1=3 \rightarrow t=1$.

Point: $(1+1, 1, 2-1) = (2,1,1)$.

Quick check · multiple choice

A plane has normal vector $\mathbf{n}=(2,-1,3)$. Which vector could represent the direction of a line perpendicular to this plane?

A $(1,2,1)$

B $(2,-1,3)$

C $(-2,1,-3)$

D Both B and C

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Line L_1 has equation $\mathbf{r}=(2,1,0)+s(1,-1,2)$. Plane Π has equation $2x+y-z=5$. [8]

(a) Show that L_1 does not lie in Π by checking whether the direction vector of L_1 is perpendicular to the normal of Π , and finding the intersection point if one exists.

(b) Find the angle between L_1 and the normal to Π , then hence find the angle between L_1 and the plane Π , to the nearest degree.

Final answer: _____



Answers and explanation

Question 1 [8]

(a) Direction of L_1 : $(1, -1, 2)$; normal of Π : $(2, 1, -1)$. Dot product = $2(1) + 1(-1) + (-1)(2) = 2 - 1 - 2 = -1 \neq 0$, so L_1 is not parallel to Π (not lying in or parallel to it), and must intersect it. Substituting $x = 2 + s$, $y = 1 - s$, $z = 2s$: $2(2 + s) + (1 - s) - 2s = 5 \rightarrow 4 + 2s + 1 - s - 2s = 5 \rightarrow 5 - s = 5 \rightarrow s = 0$. Intersection point: $(2, 1, 0)$.

(b) $\cos \theta = |(-1)| / (|(1, -1, 2)| \cdot |(2, 1, -1)|) = 1 / (\sqrt{6} \cdot \sqrt{6}) = 1/6$. $\theta = \arccos(1/6) = 80.4^\circ$. This is the angle between the line and the normal; angle between line and plane = $90^\circ - 80.4^\circ = 9.6^\circ \approx 10^\circ$.

Quick check answer

D. A line perpendicular to the plane must be parallel to the normal vector, so any scalar multiple of n (including $-n$) works.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.